Malitha Gunawardhana

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Summary

Machine Learning, Data Science, and Software Engineering professional with 5+ years of experience developing applied AI and data-driven solutions across multiple domains. Experienced in Python, deep learning, computer vision, LLMs, RAG, data pipelines, MLOps, APIs, and production AI workflows. Strong background in building reproducible machine-learning systems, evaluating model reliability, and automating complex technical workflows. Skilled at translating ambiguous requirements into practical solutions and collaborating across technical and multidisciplinary teams.

Experience

Data Processing Technician, Part-Time — March 2026 – Present
Viotel Pty Ltd — *Parnell, Auckland*

- Deployed a deep-learning based system for **seismic wave phase identification** across **3,700+ seismic events and 30,000+ station samples**, reducing manual waveform-processing time by **more than 70%**.
- Built **Python-based time-series anomaly-detection pipelines** to identify abnormal vibration patterns across **8+ telecommunications towers**, supporting automated monitoring and investigation of sensor data.
- Collaborated with geophysicists and a three-member field team to investigate sensor and data-quality issues, validate model outputs, and support deployment of **17 seismic sensors** across the Taupō region.

Machine Learning Engineer — Sep. 2022 – Jan. 2025
Institute of Fundamental Technological Research, Polish Academy of Sciences (IPPT PAN) — *Warsaw, Poland*

- Developed and evaluated deep-learning models for **breast tumour identification from ultrasound images**, contributing to a low-cost computer-aided detection system.
- Designed **reusable and reproducible medical-imaging pipelines** spanning data sourcing, curation, preprocessing, quality assurance, model training, evaluation, and cross-dataset comparison.
- Collaborated with **clinicians, researchers, and engineers** to define requirements, evaluate model behaviour, communicate limitations, and ensure technical outputs remained useful to multidisciplinary users.

Artificial Intelligence Researcher — Sep. 2022 – Sep. 2023
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — *Abu Dhabi, UAE*

- Developed and evaluated **self-supervised and semi-supervised learning** approaches for video analysis, assessing model performance under limited labelled-data conditions.
- Investigated **LLMs, network calibration, and deep-learning reliability** through controlled experiments, quantitative evaluation, and systematic analysis of model limitations and failure modes.
- Led and collaborated within a four-member research team to design experiments, analyse results, and communicate technical findings through leading artificial intelligence conferences.

Machine Learning Engineer — June 2022 – Nov. 2022
PromiseQ GmbH — *Berlin, Germany*

- Improved deep-learning models supporting a production surveillance platform operating across **3,000+ CCTV cameras**, analysing recurring failure modes and production prediction errors.
- Reduced false alarms by **5%** through network calibration, quantitative evaluation, and error analysis, improving the reliability of an AI tool used in daily operations.
- Collaborated with cross-functional teams to monitor deployed-model performance, diagnose production issues, communicate findings, and iteratively improve system reliability.

Full-Stack Software Engineer — March 2021 – May 2022
Xeptagon (Pvt) Ltd — *Colombo, Sri Lanka*

- Worked directly with customers to understand operational processes and translate ambiguous, changing requirements into production backend services and APIs.
- Led a four-member engineering team delivering a **domain drop-catching system** with over **90% success** and less than **50 ms latency**, integrating concurrent backend components and optimising execution performance.
- Built audio-processing functionality for a student learning management system, transforming **80+ audio signals** into reusable structured features through signal processing and feature engineering.

Software Engineer (Intern) — June 2019 – Dec. 2019
Synergen Technology Labs (Pvt) Ltd — *Colombo, Sri Lanka*

- Designed and supported development of a wearable physiological-signal acquisition device and collection of an original dataset through controlled stress-inducing experiments.
- Developed physiological signal-processing and ML pipelines for stress classification, extracting structured features from sensor data and achieving over **90%** accuracy across four stress categories.

Education

Doctor of Philosophy Dec. 2023 – Aug. 2026
University of Auckland *Auckland, New Zealand*

- Thesis: Robust, Uncertainty-Aware, and Scalable Deep Learning for Bi-Atrial Segmentation from LGE-MRI in Atrial Fibrillation.
- **Award: Health Research Council Scholarship.**
- Developed and evaluated **computer-vision and deep-learning models** across medical-imaging datasets, with emphasis on robustness, generalisation, uncertainty-aware evaluation, comparative performance, and reproducibility.
- Built reproducible research ML pipelines spanning data curation, preprocessing, model training, experiment tracking, statistical evaluation, cross-dataset validation, and systematic failure analysis.
- Built a **RAG-based literature agent** over a local corpus of downloaded research papers using PDF parsing, chunking, embeddings, vector search, and an LLM agent to answer natural-language queries with cited source passages, reducing literature-search time during thesis work.
- Published and presented research outcomes internationally, translating quantitative evidence into clear technical findings and conclusions.

B.Sc. Engineering Honours Jan. 2017 – July 2021
University of Moratuwa *Moratuwa, Sri Lanka*

- Dean's list placement in Semester 7.
- Key modules: Statistics, Image Processing and Machine Vision, Neural Networks and Fuzzy Logic, Medical Imaging, and Signal Processing.

Certifications & Training

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional
  * Fundamentals of LLMs, model pre-training, fine-tuning and alignment, generative AI, and MLOps
- Microsoft Certified: Azure Fundamentals (AZ-900)
  * Azure cloud services, cloud computing concepts, storage, networking, security, and virtualisation
- Large Language Model Agents, UC Berkeley
  * LLM agents, autonomous AI systems, multi-agent coordination, prompt engineering, RAG, and MLOps
- Deep Learning Specialization, DeepLearning.AI
  * Neural networks, deep learning, convolutional neural networks, and recurrent neural networks
- Data Science Career Track, 365 Data Science
  * Python, SQL, data wrangling, statistical analysis, machine learning, and AI model evaluation

Technical Skills

Programming Languages: Python (Pandas, NumPy), SQL, MATLAB, Go, TypeScript, Bash, R
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Computer Vision, Deep Learning, Time-Series Modelling
Generative AI and LLMs: RAG, LangChain, CrewAI, Weaviate, LLM Agents, Agentic Workflows, Prompt Engineering
Data and MLOps: MLflow, Apache Airflow, PySpark, Weights & Biases, Neptune, Docker, Jenkins CI/CD
Cloud and Engineering: AWS, Azure, GCP, Git, Linux, REST APIs, MongoDB, Jira, Confluence
Data Analytics and BI: Pandas, NumPy, Power BI, Microsoft Excel, Statistical Analysis, Data Visualisation

Service and Leadership

Auckland Bioengineering Institute, University of Auckland Feb. 2024 – Aug. 2026

- Elected Student Council Member representing postgraduate students in university discussions, student advocacy, welfare initiatives, and policy discussions.

Sustainable Education Foundation July 2024 – Apr. 2025

- Mentored students through structured academic, professional, and personal development guidance.

Rotaract Club of Alumni of the University of Moratuwa July 2022 – July 2024

- Served as Vice President for Club Service, coordinating member engagement, fellowship activities, and community initiatives.

References

Available upon request.